EJERCICIOS BLOQUE 3 nivel básico

Ejercicios de red (1–12)

Bloque 1: Información del sistema y red (1–8)

Ejercicio 1. Información del sistema

```
PS C:\Windows\System32> Get-ComputerInfo | Select-Object CsName, WindowsVersion, OsArchitecture

CsName          WindowsVersion OsArchitecture
------          -------------- --------------
DESKTOP-M7CM9BE 2009           64 bits

PS C:\Windows\System32>
```

Muestra el nombre del equipo, usuario actual y sistema operativo.

Get-ComputerInfo | Select-Object CsName, WindowsVersion, OsArchitecture

Ejercicio 2. Adaptadores de red

```
PS C:\Windows\System32> Get-NetAdapter | Select-Object Name, Status

Name            Status
----            ------
vEthernet (WSL) Up
Ethernet 3      Up
Ethernet 2      Up

PS C:\Windows\System32>
```

Lista los adaptadores de red mostrando nombre y estado.

Get-NetAdapter | Select-Object Name, Status

Ejercicio 3. Dirección IP local

```
PS C:\Windows\System32> Get-NetIPAddress -AddressFamily IPv4 | Where-Object {$_.IPAddress -notlike "169.*"} | Select-Object Inter
faceAlias, IPAddress

InterfaceAlias              IPAddress
--------------              ---------
vEthernet (WSL)             172.22.176.1
Ethernet 3                  192.168.0.75
Ethernet 2                  192.168.56.1
Loopback Pseudo-Interface 1 127.0.0.1

PS C:\Windows\System32>
```

Muestra la dirección IPv4 principal del equipo.

Get-NetIPAddress -AddressFamily IPv4 | Where-Object {$_.IPAddress -notlike "169.*"} | Select-Object InterfaceAlias, IPAddress

Ejercicio 4. Ping básico

```
PS C:\Windows\System32> Test-Connection google.com -Count 4

   Destination: google.com

Ping Source           Address                  Latency BufferSize Status
                                                  (ms)        (B)
---- ------           -------                  ------- ---------- ------
   1 DESKTOP-M7CM9BE  216.58.215.142               11         32 Success
   2 DESKTOP-M7CM9BE  216.58.215.142               11         32 Success
   3 DESKTOP-M7CM9BE  216.58.215.142               11         32 Success
   4 DESKTOP-M7CM9BE  216.58.215.142               11         32 Success

PS C:\Windows\System32>
```

Envía 4 pings a google.com.

Test-Connection google.com -Count 4

Ejercicio 5. Ping a varios hosts

```
PS C:\Windows\System32> $hosts = "google.com","microsoft.com","github.com"
>> foreach ($h in $hosts) { Test-Connection $h -Count 2 | Select-Object Address, ResponseTime }
>>

Address        ResponseTime
-------        ------------
216.58.215.142
216.58.215.142


140.82.121.3
140.82.121.3

PS C:\Windows\System32>
```

Haz ping a google.com, microsoft.com y github.com.

$hosts = "google.com","microsoft.com","github.com"

foreach ($h in $hosts) { Test-Connection $h -Count 2 | Select-Object Address, ResponseTime }

Ejercicio 6. Comprobar puerto abierto

```
PS C:\Windows\System32> Test-NetConnection -ComputerName www.microsoft.com -Port 80

ComputerName     : www.microsoft.com
RemoteAddress    : 23.1.250.96
RemotePort       : 80
InterfaceAlias   : Ethernet 3
SourceAddress    : 192.168.0.75
TcpTestSucceeded : True

PS C:\Windows\System32>
```

Comprueba si www.microsoft.com responde en el puerto 80.

Test-NetConnection -ComputerName www.microsoft.com -Port 80

Ejercicio 7. Ruta hacia un servidor (traceroute)

```
PS C:\Windows\System32> Test-NetConnection 8.8.8.8 -TraceRoute

ComputerName            : 8.8.8.8
RemoteAddress           : 8.8.8.8
InterfaceAlias          : Ethernet 3
SourceAddress           : 192.168.0.75
PingSucceeded           : True
PingReplyDetails (RTT)  : 11 ms
TraceRoute              : 192.168.0.1
                          10.187.4.161
                          80.58.120.113
                          81.45.103.41
                          0.0.0.0
                          81.46.8.94
                          176.52.253.93
                          213.140.50.43
                          192.178.110.87
                          142.251.49.55
                          8.8.8.8
```

Muestra la ruta hasta 8.8.8.8.

Test-NetConnection 8.8.8.8 -TraceRoute

Ejercicio 8. Guardar resultados de ping

```
PS C:\Windows\System32> Test-Connection 8.8.8.8 -Count 4 | Out-File "$env:USERPROFILE\Desktop\ping.txt"
PS C:\Windows\System32> cat $env:USERPROFILE\Desktop\ping.txt

   Destination: 8.8.8.8

Ping Source           Address                  Latency BufferSize Status
                                                  (ms)        (B)
---- ------           -------                  ------- ---------- ------
   1 DESKTOP-M7CM9BE  8.8.8.8                       10         32 Success
   2 DESKTOP-M7CM9BE  8.8.8.8                       10         32 Success
   3 DESKTOP-M7CM9BE  8.8.8.8                       10         32 Success
   4 DESKTOP-M7CM9BE  8.8.8.8                       10         32 Success

PS C:\Windows\System32>
```

Haz ping a 8.8.8.8 y guarda el resultado en un archivo de texto.

Test-Connection 8.8.8.8 -Count 4 | Out-File "$env:USERPROFILE\Desktop\ping.txt"

- **Bloque 2: Exploración del sistema y archivos (9–12)**

Ejercicio 9. Listar procesos activos

```
PS C:\Windows\System32> Get-Process | Sort-Object WS -Descending | Select-Object -First 10 Name, ID, WS

Name                  Id         WS
----                  --         --
VirtualBoxVM       10072 8061579264
eclipse            13908  785002496
Memory Compression  2144  660647936
WINWORD              368  321789952
firefox            12796  296357888
MsMpEng            13728  247566336
firefox            11528  220082176
explorer           11200  215531520
pwsh               13952  193912832
firefox            12692  165888000

PS C:\Windows\System32>
```

Muestra los 10 primeros procesos con más uso de memoria.

Get-Process | Sort-Object WS -Descending | Select-Object -First 10 Name, ID, WS

Ejercicio 10. Servicios en ejecución

```
PS C:\Windows\System32> Get-Service | Where-Object Status -eq "Running" | Select-Object Name, DisplayName

Name                      DisplayName
----                      -----------
Appinfo                   Información de la aplicación
AppMgmt                   Administración de aplicaciones
AudioEndpointBuilder      Compilador de extremo de audio de Windows
Audiosrv                  Audio de Windows
BFE                       Motor de filtrado de base
BrokerInfrastructure      Servicio de infraestructura de tareas en segundo plano
BthAvctpSvc               Servicio AVCTP
cbdhsvc_9ac04efd          Servicio de usuario del portapapeles_9ac04efd
CDPSvc                    Servicio de plataforma de dispositivos conectados
CDPUserSvc_9ac04efd       Servicio de usuario de plataforma de dispositivos conectados_9ac04efd
ClickToRunSvc             Microsoft Office Click-to-Run Service
CoreMessagingRegistrar    CoreMessaging
cphs                      Intel(R) Content Protection HECI Service
cplspcon                  Intel(R) Content Protection HDCP Service
CryptSvc                  Servicios de cifrado
DcomLaunch                Iniciador de procesos de servidor DCOM
DeviceAssociationService  Servicio de asociación de dispositivos
Dhcp                      Cliente DHCP
DiagTrack                 Experiencias del usuario y telemetría asociadas
DispBrokerDesktopSvc      Mostrar el servicio de directivas
DisplayEnhancementService Servicio de mejora de visualización
```

Muestra los servicios que están en estado “Running”.

Get-Service | Where-Object Status -eq "Running" | Select-Object Name, DisplayName

Ejercicio 11. Espacio en disco

```
PS C:\Windows\System32> Get-PSDrive -PSProvider FileSystem | Select-Object Name, Used, Free

Name          Used        Free
----          ----        ----
C    227216367616 12129800192
Temp 227216367616 12129800192

PS C:\Windows\System32>
```

Muestra las unidades disponibles y el espacio libre.

Get-PSDrive -PSProvider FileSystem | Select-Object Name, Used, Free

Ejercicio 12. Listar archivos de una carpeta

```
PS C:\Windows\System32> $ruta = Read-Host "Introduce una ruta"
>> Get-ChildItem $ruta
>>
Introduce una ruta: c:

    Directory: C:\Windows\System32

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d----          29/08/2025     9:27                %userprofile%
d----          07/12/2019    15:55                0409
d----          04/12/2023     3:52                AdvancedInstallers
d----          07/12/2019    10:14                am-et
d----          07/12/2019    10:14                AppLocker
d----          27/08/2025    13:19                appraiser
d---s          27/08/2025    13:19                AppV
d----          04/12/2023     3:52                ar-SA
d----          04/12/2023     3:52                bg-BG
d----          18/09/2025    14:56                Boot
d----          07/12/2019    10:14                Bthprops
d----          14/11/2025     8:37                CatRoot
d----          14/11/2025     8:37                catroot2
d----          27/08/2025    13:19                CodeIntegrity
d----          04/12/2023     3:52                Com
```

Pide al usuario una ruta y muestra los archivos.

$ruta = Read-Host "Introduce una ruta"

Get-ChildItem $ruta

- **Bloque 3: Eventos del sistema (13–16)**

Ejercicio 13. Últimos 10 eventos del Sistema

```
PS C:\Windows\System32> Get-WinEvent -LogName System -MaxEvents 10 | Select-Object TimeCreated, Id, LevelDisplayName, ProviderNam
e

TimeCreated           Id LevelDisplayName ProviderName
-----------           -- ---------------- ------------
14/11/2025 12:00:00 6013 Información      EventLog
14/11/2025 11:00:21 7040 Información      Service Control Manager
14/11/2025 10:59:02   15 Información      Microsoft-Windows-Kernel-General
14/11/2025 10:58:59   16 Información      Microsoft-Windows-Kernel-General
14/11/2025 10:58:59   16 Información      Microsoft-Windows-Kernel-General
14/11/2025 10:58:59   16 Información      Microsoft-Windows-Kernel-General
14/11/2025 10:58:59   15 Información      Microsoft-Windows-Kernel-General
14/11/2025 10:58:55   11 Información      Microsoft-Windows-Kernel-General
14/11/2025 10:58:55   11 Información      Microsoft-Windows-Kernel-General
14/11/2025 10:58:55   15 Información      Microsoft-Windows-Kernel-General

PS C:\Windows\System32>
```

Get-WinEvent -LogName System -MaxEvents 10 | Select-Object TimeCreated, Id, LevelDisplayName, ProviderName

Ejercicio 14. Últimos errores de aplicación

Get-WinEvent -FilterHashtable @{LogName='Application'; Level=2} -MaxEvents 5

Ejercicio 15. Contar eventos recientes

```
PS C:\Windows\System32> Get-WinEvent -FilterHashtable @{LogName='Application'; Level=2} -MaxEvents 5

   ProviderName: .NET Runtime

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
14/11/2025 12:09:21           1022 Error            .NET Runtime version 4.0.30319.0 - Error al inicializar la infraestructura …
14/11/2025 11:57:51           1022 Error            .NET Runtime version 4.0.30319.0 - Error al inicializar la infraestructura …
14/11/2025 11:51:39           1022 Error            .NET Runtime version 4.0.30319.0 - Error al inicializar la infraestructura …

   ProviderName: Microsoft-Windows-Security-SPP

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
14/11/2025 7:45:05            8198 Error            Error de la activación de licencia (slui.exe) con el siguiente código:…
14/11/2025 7:44:57            8198 Error            Error de la activación de licencia (slui.exe) con el siguiente código:…

PS C:\Windows\System32>
```

(Get-WinEvent -LogName System -MaxEvents 50).Count

Ejercicio 16. Exportar eventos a CSV

```
PS C:\Windows\System32> Get-WinEvent -LogName System -MaxEvents 20 | Export-Csv "$env:USERPROFILE\Desktop\eventos.csv" -NoTypeInf
ormation
PS C:\Windows\System32> cat $env:USERPROFILE\Desktop\eventos.csv
"Message","Id","Version","Qualifiers","Level","Task","Opcode","Keywords","RecordId","ProviderName","ProviderId","LogName","Proces
sId","ThreadId","MachineName","UserId","TimeCreated","ActivityId","RelatedActivityId","ContainerLog","MatchedQueryIds","Bookmark"
,"LevelDisplayName","OpcodeDisplayName","TaskDisplayName","KeywordsDisplayNames","Properties"
"El tiempo límite del sistema es de 2089495 segundos.","6013","0","32768","4","0","0","36028797018963968","4511","EventLog",,"Sys
tem","0","0","DESKTOP-M7CM9BE",,"14/11/2025 12:00:00",,,"System","System.UInt32[]","System.Diagnostics.Eventing.Reader.EventBookm
ark","Información",,,"System.Collections.ObjectModel.ReadOnlyCollection`1[System.String]","System.Collections.Generic.List`1[Syst
em.Diagnostics.Eventing.Reader.EventProperty]"
"El tipo de inicio del servicio Servicio de transferencia inteligente en segundo plano (BITS) se cambió de inicio automático a in
icio por solicitud.","7040","0","16384","4","0","0","-9187343239835811840","4510","Service Control Manager","555908d1-a6d7-4695-8
e1e-26931d2012f4","System","932","2100","DESKTOP-M7CM9BE","S-1-5-18","14/11/2025 11:00:21",,,"System","System.UInt32[]","System.D
iagnostics.Eventing.Reader.EventBookmark","Información",,,"System.Collections.ObjectModel.ReadOnlyCollection`1[System.String]","S
ystem.Collections.Generic.List`1[System.Diagnostics.Eventing.Reader.EventProperty]"
"El subárbol \??\Volume{6ccb0593-7878-4fc2-a6a5-637a7b7b20f3}\System Volume Information\SPP\SppCbsHiveStore\{cd42efe1-f6f1-427c-b
004-033192c625a4}{ED12B9BF-59C1-410C-9B62-C39161988137} se reorganizó con un tamaño inicial de 98168832 bytes y un tamaño final d
e 97951744 bytes.","15","0",,"4","10","0","-9223372036854775808","4509","Microsoft-Windows-Kernel-General","a68ca8b7-004f-d7b6-a6
98-07e2de0f1f5d","System","5384","5924","DESKTOP-M7CM9BE","S-1-5-18","14/11/2025 10:59:02",,,"System","System.UInt32[]","System.D
iagnostics.Eventing.Reader.EventBookmark","Información","Información",,"System.Collections.ObjectModel.ReadOnlyCollection`1[Syste
m.String]","System.Collections.Generic.List`1[System.Diagnostics.Eventing.Reader.EventProperty]"
"El historial de accesos en el subárbol \??\Volume{6ccb0593-7878-4fc2-a6a5-637a7b7b20f3}\System Volume Information\SPP\SppCbsHive
Store\{cd42efe1-f6f1-427c-b004-033192c625a4}{2D427380-2318-418E-A069-BFAC67131DA0} se borró mediante la actualización de 551 clav
es y la creación de 70 páginas modificadas.","16","0",,"4","0","0","-9223372036854775808","4508","Microsoft-Windows-Kernel-Genera
l","a68ca8b7-004f-d7b6-a698-07e2de0f1f5d","System","5384","5924","DESKTOP-M7CM9BE","S-1-5-18","14/11/2025 10:58:59",,,"System","S
```

Get-WinEvent -LogName System -MaxEvents 20 | Export-Csv "$env:USERPROFILE\Desktop\eventos.csv" -NoTypeInformation

- **Bloque 4: Tareas y automatización (17–20)**

Ejercicio 17. Listar tareas programadas

```
PS C:\Windows\System32> Get-ScheduledTask | Select-Object TaskName, State

TaskName                                                                    State
--------                                                                    -----
AbrirNotepad                                                                Ready
MicrosoftEdgeUpdateTaskMachineCore                                          Ready
MicrosoftEdgeUpdateTaskMachineUA                                            Ready
MSI_Driver_Utility_Installer                                                Ready
npcapwatchdog                                                               Ready
OneDrive Standalone Update Task-S-1-5-21-374513637-3669074883-229338491-1004 Ready
DropboxUpdaterTaskSystem123.0.6299.129{89CC420A-1240-43E3-BE4C-EA27A84F1D1F} Ready
GoogleUpdaterTaskSystem141.0.7376.0{884A0BE0-690E-4CD5-BCDF-3E25BD07A5F1}   Ready
Office Actions Server                                                       Ready
Office Automatic Updates 2.0                                                Ready
Office Background Push Maintenance                                          Ready
Office ClickToRun Service Monitor                                           Ready
Office Feature Updates                                                      Ready
Office Feature Updates Logon                                                Ready
Office Performance Monitor                                                  Ready
.NET Framework NGEN v4.0.30319                                              Ready
.NET Framework NGEN v4.0.30319 64                                           Ready
.NET Framework NGEN v4.0.30319 64 Critical                                Disabled
```

Get-ScheduledTask | Select-Object TaskName, State

Ejercicio 18. Crear tarea para abrir el bloc de notas

```
PS C:\Windows\System32> $accion = New-ScheduledTaskAction -Execute "notepad.exe"
>> $trigger = New-ScheduledTaskTrigger -Once -At (Get-Date).AddMinutes(1)
>> Register-ScheduledTask -TaskName "AbrirNotepad2" -Action $accion -Trigger $trigger
>>

TaskPath                                       TaskName                       State
--------                                       --------                       -----
\                                              AbrirNotepad2                  Ready

PS C:\Windows\System32>
```

$accion = New-ScheduledTaskAction -Execute "notepad.exe"

$trigger = New-ScheduledTaskTrigger -Once -At (Get-Date).AddMinutes(1)

Register-ScheduledTask -TaskName "AbrirNotepad" -Action $accion -Trigger $trigger

Ejercicio 19. Ejecutar manualmente una tarea

```
PS C:\Windows\System32> Start-ScheduledTask -TaskName "AbrirNotepad2"
PS C:\Windows\System32> Start-ScheduledTask -TaskName "AbrirNotepad2"
PS C:\Windows\System32>
```

Start-ScheduledTask -TaskName "AbrirNotepad"

Ejercicio 20. Exportar tareas a CSV

Get-ScheduledTask | Select-Object TaskName, State | Export-Csv "$env:USERPROFILE\Desktop\tareas.csv"

```
PS C:\Windows\System32> Get-ScheduledTask | Select-Object TaskName, State | Export-Csv "$env:USERPROFILE\Desktop\tareas.csv"
PS C:\Windows\System32> cat "$env:USERPROFILE\Desktop\tareas.csv"
"TaskName","State"
"AbrirNotepad","Ready"
"AbrirNotepad2","Ready"
"MicrosoftEdgeUpdateTaskMachineCore","Ready"
"MicrosoftEdgeUpdateTaskMachineUA","Ready"
"MSI_Driver_Utility_Installer","Ready"
"npcapwatchdog","Ready"
"OneDrive Standalone Update Task-S-1-5-21-374513637-3669074883-229338491-1004","Ready"
"DropboxUpdaterTaskSystem123.0.6299.129{89CC420A-1240-43E3-BE4C-EA27A84F1D1F}","Ready"
"GoogleUpdaterTaskSystem141.0.7376.0{884A0BE0-690E-4CD5-BCDF-3E25BD07A5F1}","Ready"
"Office Actions Server","Ready"
"Office Automatic Updates 2.0","Ready"
"Office Background Push Maintenance","Ready"
"Office ClickToRun Service Monitor","Ready"
"Office Feature Updates","Ready"
"Office Feature Updates Logon","Ready"
"Office Performance Monitor","Ready"
".NET Framework NGEN v4.0.30319","Ready"
".NET Framework NGEN v4.0.30319 64","Ready"
".NET Framework NGEN v4.0.30319 64 Critical","Disabled"
".NET Framework NGEN v4.0.30319 Critical","Disabled"
"AD RMS Rights Policy Template Management (Automated)","Disabled"
"AD RMS Rights Policy Template Management (Manual)","Ready"
```